

Firstki Aditya Fernanda

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Firstki Aditya is a final-year Information Systems major with hands-on experience in penetration testing, vulnerability assessment, web security, networking, and secure application development. Experienced in conducting black-box security testing, plugin, enumeration, vulnerability validation, threat modelling, and security analysis using the OWASP Framework. Conducted thesis research across 6 virtual Moodle servers and 5 infrastructure layers, identifying and validating vulnerabilities including SQL Injection, XSS, IDOR, RCE, SSRF, CSRF, and system or container misconfigurations. Supported by BNSP Pentester certification and practical experience in secure authentication, RBAC, RESTful API security, network troubleshooting, and technical mentoring.

EDUCATION

Telkom University, Bandung, West Java. Sep 2022 - Present

Bachelor's degree in information systems

- **Lecture Assistant:** Programming Basics Laboratory, Network System Laboratory, and Enterprise Infrastructure Management Laboratory.
- **Course Related:** Web Programming Basic, Data Mining Course, and Network System Course.
- **Project:** Inventory Management System, Traction Fleet Management System, Landing Page, Bidding Website, and Human Resource Management System.
- **Research:** Analysis of Plugin Vulnerabilities in Moodle-Based E-Learning Websites via Enumeration Using the OWASP Framework

WORK EXPERIENCE

Mentor Cybersecurity, *Contract*. December 2024 – February 2025

Jagoan Hosting and Practiclass, Remote, Indonesia.

- Mentored participants on **cybersecurity fundamentals, web security, vulnerabilities, and system hardening** through hands-on exercises and practical demonstrations.
- Guided mentees in understanding **common security vulnerabilities and defensive techniques**, while providing technical support and troubleshooting.
- Translated complex cybersecurity concepts into practical explanations to improve participants' technical understanding and engagement.
- Recognized as a **"Most Favorite Instructor Nominee"** for dedication and positive impact on mentees' learning experience.

Cybersecurity Webinar Speaker, *Contract*. July 2025 – December 2025

NGALUP Collaborative Network, Remote, Indonesia.

- Delivered a cybersecurity webinar covering **Firewall & Firewall Policy, Deny-by-Default**, and security measures against **DDoS and Botnet attacks**.
- Shared practical insights and preparation strategies for **LKS (Lomba Kompetensi Siswa) cybersecurity/infrastructure competitions**.
- Educated participants on fundamental cybersecurity practices and personal protection against common cyber threats.
- Served as a speaker by leveraging experience as **Silver Medalist of LKS East Java 2022** and expertise in **Enterprise Infrastructure Management**.

Software Engineer, *Internship*. January 2025 – June 2025

PT Rasa Aksata Nusantara (Duluin Fintech). Bandung, West Java, Indonesia.

- Repaired **10+ critical production bugs** using **NodeJS**, maintaining system stability during high-traffic periods.

- Reviewed performance bottlenecks and resolved **reducing page load time by 25%**.
- Implemented **AJAX-based asynchronous data fetching**, reducing server load and improving user interaction seamlessness by **20%**.
- Optimized **SQL queries and indexing** for transactional logs, resulting in **30% faster data retrieval** for monthly report generation.
- Collaborated on implementing **RESTful API security protocols**, reducing potential vulnerability risks by ensuring **100% compliance** with internal data encryption standards.

PROJECT

Undergraduate Thesis. July 2025 – December 2025

Analysis of Plugin Vulnerabilities in Moodle-Based E-Learning Websites via Enumeration Using the OWASP Framework

- Conducted **black-box penetration testing** and **plugin enumeration** across **6 Moodle servers** and **5 infrastructure layers** using **Nmap, OWASP ZAP, Nuclei, WhatWeb, Nikto, MoodleScanner**, and other security tools.
- Identified **SQL Injection, XSS, IDOR, RCE, SSRF, CSRF**, security misconfigurations, and container vulnerabilities, including **48 public vulnerabilities** on Moodle.
- Manually validated **122 security findings** with **47 True Positives, 23 True Negatives, 0 False Positives, and 52 False Negatives**, and correlated findings with **NVD, MITRE CWE, and CIS Benchmarks**.
- Applied **OWASP, Attack Tree threat modeling, and IAAA analysis** to map attack paths and develop remediation recommendations covering **security hardening, access control, security headers, least privilege, and container security**.

Software Engineer, Contract. January 2025 – June 2025

Project Fleet Management System (TRACTION)

- Developed a **real-time tracking module using WebSocket**, enabling live updates for **50+ fleet vehicles** simultaneously.
- Optimized **database queries** for asset tracking, resulting in **30% faster retrieval** of historical trip data.
- Designed and implemented a **RESTful API with Redis caching**, reducing database load by 40% and maintaining a response time of **under 100ms** for high-frequency location pings.
- Developed a **Role-Based Access Control (RBAC)** system, ensuring secure data isolation for different user levels (**Driver, Manager, Admin**) and preventing unauthorized access to sensitive fleet logs.

ORGANIZATION

Team Lead Programming Basic Laboratory. August 2023 – January 2024

Telkom University, Bandung, West Java, Indonesia.

- Allocated task responsibilities to practicum assistants and organized a team of **6-24 assistant**.
- Lectured practical work for **4 classes** and guiding **7 to 9 student per class**.
- Delivered report regularly to the department head or faculty supervisor.
- Responded students, practicum assistant, and faculty to ensure response to queries within **48 hours**.

Competition Division Staff. July 2024 – July 2025

Enterprise Infrastructure Management Laboratory, Bandung, West Java, Indonesia.

- Collaborated with cross-functional teams to develop strategic initiatives aligned with **enterprise infrastructure objectives**.
- Conducted **research and technical analysis** on emerging **cybersecurity and infrastructure technologies**.

- Mentored students on **fundamental cybersecurity concepts**, focusing on **web security, secure configuration, and system hardening techniques**
- Promoted best practices in **secure system design and infrastructure resilience**.

Practicum Assistant. September 2024 - October 2025

Telkom University, Bandung, West Java, Indonesia.

- Led hands-on **networking laboratory sessions for 3 classes** (7-9 students per class).
- Designed and developed introductory **Computer Network modules** covering **TCP/IP fundamentals, subnetting, and cabling (crimping)**.
- Administered practical assessments to evaluate students' understanding of **network configuration and troubleshooting**.
- Maintained and updated laboratory materials and infrastructure to ensure consistent learning quality.
- Mentored students in **network troubleshooting** and **foundational security awareness** within infrastructure environments

ACHIEVEMENT

- [2024] 1st Place Winner AI App Development International Young Entrepreneur Society Competition.
- [2024] 2nd Place Winner CR-Youth IT by Search Telkom University.
- [2023] The Best No Code Hackathon by Garuda Hacks.
- [2023] The Best Performance Hacking Program Innovation by Agility Transformation.
- [2022] 2 Place of LKS Cyber Security Competition Region Jawa Timur Province
- [2022] 4 Place of LKS Cyber Security Competition Region WILKER 3 Jawa Timur Province

CERTIFICATION

- [2026] [[Certificate](#)] BNSP Cybersecurity Analyst
- [2026] [[Certificate](#)] HackerRank. Problem Solving (Basic & Intermediate).
- [2023] [[Certificate](#)] Cyber Academy Indonesia. Web Penetration Testing
- [2022] [[Certificate](#)] DQLab. Introduction to Data with Python.

SKILLS

- **Hard Skills:** Cybersecurity & Vulnerability Assessment, Penetration Testing, Web Security, Network Security, Security Hardening, Firewall & Firewall Policy, Threat Modeling, Security Testing, RESTful API Security, Role-Based Access Control (RBAC), Database Security, Incident Analysis, System & Infrastructure Security.
- **Soft Skills:** Analytical Thinking, Problem Solving, Technical Research, Technical Mentoring, Public Speaking, Team Collaboration, Communication, Attention to Detail.
- **Technical Skills:** Python, JavaScript (ES6+), PHP, SQL, HTML5, CSS3, Node.js, Express.js, Laravel, React.js, RESTful API, WebSocket, AJAX, MySQL, PostgreSQL, Redis, Git/GitHub, Docker, Postman, Figma, Jira.
- **Cybersecurity Tools:** Nmap, OWASP ZAP, Nuclei, WhatWeb, Nikto, MoodleScanner, Burp Suite, Wireshark, Docker, Linux.

LANGUAGES

- Bahasa Indonesia (**Native**)
- English (**Intermediate**)